

Protocol for a Systematic Review of Industrial Animal Operations and Community Health

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Abstract

Background: Meat, milk, and egg production is increasingly taking place in industrial animal operations. Although these operations generate chemical, biological, and other hazardous agents in the surrounding environment, the health risks to people living in adjacent communities are unclear.

Objective: To conduct a systematic review to determine whether a relationship exists between human health and residential exposure to operations where animals are produced for food.

Methods: A comprehensive search strategy was developed to retrieve relevant studies from Medline (Ovid), Global Health (Ovid), Embase.com, Scopus, and Web of Science databases. Title/abstract and full-text screening by two independent reviewers will be conducted to identify relevant studies. We will include primary research studies focused on humans who live near animal operations. Details regarding study design, study population, and health outcomes will be extracted. The Navigation Guide methodology will be used to conduct a risk of bias assessment. Meta-analysis for selected outcomes will be conducted if possible; other quantitative or qualitative approaches will also be utilized to summarize the outcome data.

Discussion: This systematic review will provide an updated assessment of the association between proximity to animal operations and the health of people living in adjacent communities.

Key words: animal feeding operations, community health, environmental health, epidemiology, industrial agriculture, intensive animal production, livestock, systematic review

1. Introduction

1.1 Rationale

Meat, dairy, and egg production has undergone a rapid transition towards a model characterized by high stocking densities, large-scale operations, high throughput, specialization, mechanization, and greater reliance on offsite inputs (Clay et al., 2020; Gilbert et al., 2015; Wallinga et al., 2022). The industrial production of animals for food is, in many cases, the dominant model in high-income countries and is expanding into a growing number of low- and middle-income countries (Lam et al., 2019, Gilbert et al., 2015).

A growing body of literature has assessed associations between industrial animal production and an array of adverse health effects among persons living proximate to the operations. Chemical, biological, and other hazardous agents released to surrounding communities via several exposure pathways have been hypothesized to be responsible for these effects (Miralha et al., 2022; Grzinic et al., 2023). Examined hazards include (but are not limited to) hazardous air pollutants (e.g., particulate matter, ammonia, and hydrogen sulfide), and zoonotic pathogens (e.g., avian influenza viruses, hepatitis E virus, pathogenic *Escherichia coli*, *Streptococcus suis*, *Campylobacter*, *Cryptosporidium parvum*, and methicillin-resistant *Staphylococcus aureus*) (Poulsen et al., 2018; Guo et al., 2022). The generation and spread of environmental hazards are not limited to animal housing and may additionally include land where manure is applied (Casey et al., 2013), all of which we include under the term “animal operations.”

Understanding the relationship between residential proximity to industrial animal operations and health outcomes is essential for informing public health policies, land use planning decisions, and community interventions. Previous systematic reviews assessing potential risks associated with living near animal operations (the reviews were not specific to industrial scale or methods) have reported inconclusive findings (O’Connor et al., 2010, 2017), but were documented to have significant flaws in conduct (Nachman et al., 2017), and were inclusive of literature only up to 2014. There have been numerous publications on industrial animal operations and their association with human health in neighboring communities in the years since the last published review [e.g., Kravchenko et al. (2018); Schultz et al. (2019)], offering an opportunity for an updated, rigorous examination of the evidence. By systematically synthesizing and critically evaluating the available literature, this study aims to inform evidence-based decision-making and promote the health and well-being of communities residing near operations where animals are produced for food.

99 1.2 Objective
100
101 The objective of our research is to conduct a systematic review to determine whether a
102 relationship exists between residential exposure to animal operations and human health.
103
104 **2. Methods**
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106 2.1 Eligibility criteria
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108 The Population-Exposure-Comparator-Outcome (PECO) framework ([4.6.1. Table 1](#)) will be used
109 to identify eligible studies.
110
111 2.2 Information sources
112
113 Our team includes an information specialist (L.R.) with extensive expertise in developing and
114 performing literature searches for systematic reviews. We designed database-specific search
115 terms and strategies in the following databases: Medline (Ovid), Global Health (Ovid),
116 Embase.com, Scopus, and Web of Science.
117
118 2.3 Search strategy
119
120 The search strategy, shown in [4.6.2. Table 2](#), outlines the principal search concepts of animal
121 production (search lines 1-8), community health outcomes (search lines 10-13), specific health
122 outcomes (search lines 16-33) and epidemiologic studies (search lines 35-37). The search
123 strategy builds upon a previous systematic review (O'Connor et al., 2017), with a major
124 modification being additional health outcome specific terms combined with epidemiologic
125 studies to capture a more holistic set of outcomes and a broader set of health-relevant studies.
126 The epidemiologic study concept is adapted from searches available on the InterTASC
127 Information Specialists' Sub-Group (ISSG) Search Filter Resource (ISSG, 2024), which provides
128 researchers with study design-specific search filters.
129
130 2.4 Study records
131
132 *2.4.1 Data management*
133
134 All search results will be imported into EndNote to compile references and remove duplicate
135 records. DistillerSR (Ottawa, Canada) will be used for title/abstract and full-text screening.
136 During title/abstract screening, the titles and abstracts will be evaluated using the inclusion

criteria to identify relevant studies ([Table 1](#)). Articles will be eligible for this review when they describe peer-reviewed primary research in which people living in communities in close proximity and with exposures to animal operations are compared to less- or not-exposed populations for any health outcome. Studies that pass title/abstract screening will then undergo full text review to confirm whether the reference conforms to inclusion criteria for inclusion in the final review. Studies that do not adhere to the PECO statement or inclusion/exclusion criteria in either step will be excluded from the final list of included studies.

2.4.2 Selection process

Each reference will be independently screened by two reviewers at the title/abstract screening and full-text screening steps. Disputes that cannot be resolved by discussion between the two reviewers will be handled by a third reviewer. The first 10% of screened studies will be reviewed for quality control by a third reviewer as well.

2.4.3 Data collection process

Data will be extracted from each included study by two investigators using DistillerSR. In the event of discrepancies in data extracted that cannot be resolved with discussion between the two reviewers, a third reviewer will be consulted to reach consensus. For studies that do not report all the data or are missing information (for example, blinding), we will attempt to contact the authors with an initial e-mail and one follow-up e-mail query with a limit of two weeks before the data are categorized as “missing”.

2.5 Data items

The study population size and geographic location will be recorded as well as the types of individuals recruited within the community (e.g., neighboring residents, community schools, etc.). We will extract all available demographic information on the study population including but not limited to age, sex, socioeconomic status, race and/or ethnicity. If the study provides information about the population’s baseline comorbidities, we will also record this information. Information regarding the animal population at the operation in each study will be extracted such as animal type (e.g., swine, poultry), animal population size, manure management practices, and any other descriptive information on the animal population when available. The study type will be recorded (e.g., cross-sectional, prospective cohort, retrospective cohort, etc.). The measure of exposure, and metric used to define exposure to the animal operation will be recorded as well (e.g., proximity to facility, perceived odor, endotoxin level). In addition, we will record the health outcome measure(s) and how the studied health outcome(s) were

175 assessed (e.g., pulmonary function test, physician diagnosis, self-report). Other variables that
176 will be extracted include the blinding of data collection, the funding source(s) for the study, the
177 statistical approach used to analyze the relationship between exposure and outcome, and
178 covariates used in multivariable analyses.

179
180 In the event of duplicate publications, companion documents, or multiple reports of a primary
181 study, we will collate all available data and use the most complete dataset including the highest
182 quality estimate of the exposure-outcome relationship taken from all identified publications.

183
184 2.6 Outcomes and prioritization
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186 All health-relevant outcomes will be included. [4.6.3. Table 3](#) highlights several examples of
187 anticipated study health outcomes.

188
189 2.7 Risk of bias in individual studies
190

191 This review will use the Navigation Guide methodology (Woodruff & Sutton, 2014) to assess
192 each included study's internal validity or risk of bias (RoB). The Navigation Guide methodology
193 draws on the approaches used by the U.S. Environmental Protection Agency and the
194 International Agency for Research on Cancer, as well as the Grading of Recommendations,
195 Assessment, Development, and Evaluation (GRADE) approach. The Navigation Guide is
196 specifically designed to account for reliance on human observational studies instead of
197 randomized-controlled trials. The Navigation Guide assesses studies according to nine domains:
198 source population representation, blinding, exposure assessment, outcome assessment,
199 incomplete outcome data, selective outcome representation, confounding, conflict of interest,
200 and other bias (Eick et al., 2020). Each domain is assessed and rated individually, without
201 providing an overall risk of bias rating for each study.

202
203 We will utilize Health Assessment Workspace Collaborative (HAWC), a health assessment
204 management software designed by the U.S. Environmental Protection Agency used to rate and
205 visualize RoB ratings (US Environmental Protection Agency [EPA], 2021). Each study will
206 undergo RoB assessment by two co-investigators. The HAWC visualization tool will be used to
207 display the findings of the RoB assessment.

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209
210 2.8 Data synthesis
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213 Based on previous reviews (O'Connor et al., 2010, 2017), we anticipate that many of the studies
214 identified in our review will have heterogeneity in their design, such as exposure assessment,
215 outcome assessment, and statistical methods. Given the aim of summarizing as much available
216 evidence on the topic as possible, we will not exclude studies solely on the basis of compatibility
217 with other studies for meta-analysis but aim to combine compatible studies when possible, to
218 generate summary estimates that could provide insight into quantitative relationships between
219 residential exposure to animal operations and adverse impacts on human health.

220
221 With this in mind, we will quantitatively synthesize data into summary measures when the
222 individual studies are sufficiently similar, allowing for meaningful comparisons and synthesis of
223 findings. This would include considerations such as whether study designs are compatible (e.g.,
224 all cross-sectional design), whether exposures and outcome measures are comparable, where
225 outcome measures are reported on similar scales or can be readily converted to the same scale,
226 and when the included studies are methodologically sound. Potential summary statistics may
227 include prevalence Odds Ratios (OR), regression coefficients (β s, or mean differences), or
228 prevalence ratios. Heterogeneity will be explored using I^2 statistic, a measure that quantifies the
229 degree of variability among effect sizes from individual studies. It represents the proportion of
230 total variation across studies that is due to heterogeneity rather than chance. The I^2 statistic is
231 interpreted as indicating low heterogeneity (0-40%), moderate heterogeneity (30-60%),
232 substantial heterogeneity (50-90%), or considerable heterogeneity (75-100%) (Higgins et al.,
233 2023). Although an assessment of the I^2 can be informative, it is also important to contextualize
234 this metric with other factors such as study quality, sample size, and methodological diversity
235 across studies when interpreting the meta-analysis results.

236
237 We may also perform sensitivity analyses by examining the effects of excluding studies with
238 heterogeneous results as well as performing subgroup analyses based on excluding individual or
239 subsets of studies with characteristics that might be influential. If enough studies are available,
240 a more formal approach to perform a subgroup analysis, clustering studies by specific
241 characteristics to determine the impact on statistical heterogeneity, will be conducted.

242
243 For studies that are not amenable to quantitative synthesis, we will use the Synthesis without
244 Meta-analysis (SwiM) reporting guideline, which was developed for use in systematic reviews in
245 which meta-analyses of effect estimates may not be possible or appropriate for at least some
246 outcomes (Campbell et al., 2020). Although this guideline was developed for the presentation
247 of intervention studies rather than observational studies, many of the reporting items outlined
248 are still applicable to observational study presentation including the need to explain criteria
249 used to prioritize results for summary and synthesis, explain and provide a rationale for groups
250 used in synthesis, describe methods used to assess certainty of the synthesis findings, describe

data presentation methods, and to report on the limitations of the synthesis (Campbell et al., 2020)

2.9 Meta-bias(es):

We plan to assess publication bias if a sufficient number of studies is available. An evaluation of publication bias is usually recommended when there are more than ten studies, as with fewer studies these approaches may be unreliable (Dalton et al., 2017). Techniques such as Egger's regression (Egger et al., 1997) or symmetry of funnel plots (Sterne & Egger, 2001) will be utilized in combination with an assessment of the nature of the observed publication bias (i.e., assessing the potential reasons for an asymmetric funnel plot).

2.10 Confidence in cumulative evidence:

Upon completion of risk of bias assessment and quantitative/qualitative data analysis, we will use the Navigation Guide methodology to rate the overall quality and strength of the evidence. Possible ratings for the overall quality of evidence are "high," "moderate," or "low." The initial quality level of human observational data will be assumed to start at "moderate," following the framework of past Navigation Guide case studies (Johnson et al., 2014; Lam et al., 2016, 2017, 2021). Factors that may decrease the quality level of the overall body of evidence include:

1. Risk of Bias Across Studies: Study limitations – a substantial risk of bias across body of evidence;
2. Indirectness: Evidence was not directly comparable to the question of interest (i.e., population, exposure, comparator, and outcome).
3. Inconsistency: Widely different estimates of effect (heterogeneity or variability in results);
4. Imprecision: Studies had few participants and few events (wide confidence intervals); and
5. Publication Bias: Studies missing from body of evidence, resulting in an overestimate or underestimate of true effects from exposure.

Factors that may increase the quality level of the overall body of evidence include:

1. Large magnitude of effect: Upgraded if modeling suggested confounding alone unlikely to explain associations with large magnitude of effect.
2. Dose-response: Upgraded if consistent dose response gradient in one or multiple studies, and/or dose response across studies.

3. Residual confounding increases confidence: Upgraded if consideration of all plausible residual confounders, biases, or effect modification would underestimate the effect or suggest a spurious effect when results show no effect.

Study authors evaluating the quality of evidence will document their rationale supporting their ratings and all factors that contributed to their final decision. Study authors will independently evaluate the quality of evidence, then gather to compare their evaluations, discuss any discrepancies, and resolve any differences through discussion until consensus is reached, if possible. The rationale for each decision overall will be recorded. Lack of consensus for the rating on any specific factor does not preclude consensus on the overall quality of the evidence.

Subsequent to rating the quality of the evidence, review authors will rate the strength of the evidence. The overall strength of the evidence is based on a combination of four criteria: 1) quality of body of evidence (i.e., the rating from the previous step); 2) direction of effect; 3) confidence in effect; and 4) any other compelling attributes of the data that may influence certainty. The results of rating the strength of the human evidence will then be compared to the criteria specified in the Navigation Guide methodology for one of the following concluding statements: 1. Sufficient; 2. Limited; 3. Inadequate; or 4. Evidence of lack of toxicity. Any discrepancies between the reviewers' decisions will be resolved through discussion. The senior author (K.N.) will be the ultimate arbiter of any discrepancies that cannot be resolved through consensus among the review authors. All decisions, discrepancies, and discussion will be documented as part of this process and included in the manuscript to be submitted for peer-review publication.

3. Discussion

Much of the relevant literature has focused on health effects associated with animal operations that could reasonably be characterized as industrial (e.g., Quist et al., 2022). Operations classified as "large concentrated animal feeding operations" (CAFOs) in the United States may house, for example, upward of 125,000 chickens or 10,000 pigs (US EPA, n.d.). Outside countries with such clear designations, however, the ability to classify an operation as "industrial" would require either enough information to apply objective criteria (e.g., based on animal stocking densities), or subjective use and interpretations of terminology (e.g., whether operations described as "intensive" or "large" would meet the criteria for "industrial"). Given these challenges, we chose search terms and inclusion criteria that are not limited to specific scales or methods of production, as in prior reviews (O'Connor et al., 2010, 2017). If following screening the selected manuscripts provide sufficient information to parse out findings specific to industrial operations, we will do so in a *post hoc* analysis.

We anticipate considerable variation across studies in methods use for characterization of exposure to animal operations; methods such as self-report, aggregation to a specified geographical area or location, distance-based methods, interpolation, direct environmental sampling of indicator contaminants, and others have been used in the literature (Casey et al., 2015). Despite this, by categorizing studies with similar exposure methods and common health outcomes, quantitative data synthesis across studies (i.e. meta-analyses) may still be possible. If not, we anticipate there will still be value in qualitative synthesis of the weight of evidence in determining whether a relationship exists between residential proximity to animal operations and a particular health outcome.

In addition, we expect that some studies will not provide detailed information on the properties of the operations (i.e., type and number of livestock, size of the facility, methods for storage and handling of waste, etc.) under study, which by introducing non-systematic error in exposure assessment may tend to bias estimates of associations towards the null. If such a phenomenon were to occur, individual studies included in our sample may be less likely to detect a true association were it to exist.

Also, some study populations may include participants who work in animal operations and have occupational exposures in addition to residential exposures without specifying the inclusion of such persons. Such an inclusion absent explicit characterization of occupational exposures may bias effect estimates in an unpredictable fashion.

4. Other Information

4.1 Dictionary

Health will be defined according to the Constitution of the World Health Organization as a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity (World Health Organization, 2024).

Animal operations will be defined to include operations where animals are produced for food, and include animal housing, manure storage, and land where manure is applied.

Residential proximity will be defined as close physical proximity to an animal operation.

4.2 CRediT statement

363 Richard D. Semba: Methodology, Writing – Original Draft, Writing – Review & Editing,
364 Conceptualization, Supervision
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371 Lori Rosman: Methodology, Writing – Review & Editing
372 Keeve E. Nachman: Methodology, Writing - Original Draft, Writing – Review & Editing,
373 Conceptualization, Supervision, Project Administration
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384
385 4.5. References
386
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 502

503 4.6 Supplementary Information
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505 4.6.1. Table 1. Eligibility Criteria Using the PECO Framework
 506

	Inclusion Criteria	Exclusion Criteria
Population	People living in the community close to animal operations	workers on animal operations (i.e., occupational health studies)

Exposure	Exposure to an animal operation estimated/assessed using any relevant metric (e.g., distance to the nearest production site, endotoxin level, perceived odor)	Studies without an exposure to an animal operation
Comparator	Must have a comparator group with different level of/reduced exposure to variable exposure (e.g. either less exposed or not exposed at all) to intensive animal production	Studies without a comparator group
Outcome	Any adverse health outcome	N/A
Study Design	- Case-control - Cohort - Cross-sectional - Longitudinal - Ecological	- Randomized controlled trials - Systematic reviews - Meta analyses
Types of Literature	Peer-reviewed literature	- Gray literature - Trial registers - Meeting posters - Meeting abstracts - Case reports
Language of Publication	All languages	N/A

4.6.2. Table 2. Proposed Search Strategy for Medline (Ovid)

1.	Animal Husbandry/ or Housing, Animal/
2.	(([animal* or bovine or cow or cows or cattle or beef or steer or pig or pigs or piglet* or pork or swine or porcine or hog or hogs or finisher* or sheep or lamb or lambs or mutton or goat or goats or poultry or chicken* or hen or hens or broiler* or turkey* or dairy or livestock or live stock or intensiv* or industrial* or confined or confinement or concentrated or large-scale] adj4 (feed* operation* or feed* facilit*)).ti,ab,kf.
3.	(cafo or cafos).ti,ab,kf.

4. (feed lot* or feedlot* or feedyard* or feed yard*).ti,ab,kf.
5. ((animal* or bovine or cow or cows or cattle or beef or steer or pig or pigs or piglet* or pork or swine or porcine or hog or hogs or finisher* or sheep or lamb or lambs or mutton or goat or goats or poultry or chicken* or hen or hens or broiler* or turkey* or dairy or livestock or live stock) adj (operation* or facility or facilities or confined or confinement or density)).ti,ab,kf.
6. ((confined or confinement) adj3 (feed or feeding)).ti,ab,kf.
7. ((intensive or intensively or large-scale or industrial) adj3 (farm or farms or farming or livestock or live stock or animal operation*)).ti,ab,kf.
8. ((animal production or food animal production or livestock production or live stock production) adj (operation* or facility or facilities or industrial or intensiv* or high density)).ti,ab,kf.
9. or/1-8
10. Public Health/ or Environmental Health/ or Environmental Exposure/ or Inhalation Exposure/ or Environmental Pollutants/ or exp Air Pollutants/ or Water Pollutants/ or Environmental Illness/ or Environmental Monitoring/ or exp Population Density/
11. (public health* or environmental health* or environmental medicine or community health* or environmental reservoir*).ti,ab,kf,jn,jw.
12. ((public or community or communities or resident* or residence* or neighbor* or neighbour* or family or families or local* or population* or populace* or school* or preschool* or highschool* or nursery or nurseries or playgroup* or play group* or kindergarten* or student* or day care* or daycare* or "zip code*" or census) adj5 (health or disease* or impact* or effect* or exposure* or expose* or outcome* or symptom* or risk*)).ti,ab,kf.
13. ((public or community or communities or resident* or residence* or living or neighbor* or neighbour* or family or families or local* or population* or populace* or school* or preschool* or highschool* or nursery or nurseries or playgroup* or play group* or kindergarten* or student* or day care* or daycare* or "zip code*" or census) adj5 (proximity or vicinity or location* or located or nearby or near or close or closely or area* or region* or density or densities or neighboring)).ti,ab,kf.
14. or/10-13
15. 9 and 14
16. exp Exanthema/ or exp Eczema/ or exp Dermatitis, Atopic/ or exp Eye Burns/ or exp Conjunctivitis/ or (rash or rashes or exanthema* or (skin adj2 irritat*) or eczema* or (dermatit* adj2 atopic) or (neurodermatit* adj2 atopic) or (burn* adj2 eye*) or (water* adj2 eye*) or conjunctivitis).ti,ab,kf.
17. "Signs and Symptoms, Respiratory"/ or exp Cough/ or exp Respiratory Sounds/ or exp Rhinitis/ or exp Pharyngitis/ or exp Rhinitis, Allergic, Seasonal/ or exp Respiratory Hypersensitivity/ or exp Dyspnea/ or exp Sneezing/ or (respiratory or cough* or wheez* or rhiniti* or "sore throat*" or "hay fever*" or allerg* or dyspnea* or (short* adj2 breath*) or sneez* or "nasal congest*").ti,ab,kf.
18. exp Immunity, Mucosal/ or exp Immunoglobulin A, Secretory/ or ("mucosal immune function*" or "mucosal immunity" or "secretory immunoglobulin*" or "secretory IgA" or SigA or "exocrine IgA" or "colostral IgA").ti,ab,kf.
19. exp Methicillin-Resistant Staphylococcus aureus/ or exp Clostridioides difficile/ or exp Enterococcus/ or exp Campylobacter/ or exp Campylobacter Infections/ or exp Psittacosis/ or exp Salmonella Infections/ or ("Methicillin-Resistant Staphylococcus aureus" or MRSA or "nasal carriage*" or "nasal colonization" or "Q fever*" or "Clostridioides difficile" or "Clostridium difficile" or "c. diff*" or Enterococcus or Campylobacter or "Campylobacter Infection*" or campylobacterios* or psittacos* or ornithos* or "salmonella infection*" or salmonellos*).ti,ab,kf.
20. exp Stillbirth/ or exp Birth Weight/ or exp Premature Birth/ or exp Infant, Premature/ or exp Fetal Growth Retardation/ or ("birth outcome*" or stillbirth* or "birth weight*" or birthweight* or (prematur* adj2 birth*) or (preterm* adj2 birth*) or (prematur* adj2 infant*) or (preterm* adj2 infant*) or (neonat* adj2 prematur*) or (growth adj2 retard*) or (growth adj2 restrict*) or (small adj2 "gestational age*") or LBW or ELBW or SGA).ti,ab,kf.
21. exp Respiratory Physiological Phenomena/ or exp Respiratory Function Tests/ or ("lung function*" or "pulmonary function*" or "forced expiratory volume" or FEV or FWV1 or "forced vital capacity" or FVC or "Tiffeneau index" or "Tiffeneau-Pinelli index" or spiometr*).ti,ab,kf.

22. exp Respiratory Tract Diseases/ or exp Asthma/ or exp Pulmonary Disease, Chronic Obstructive/ or exp Pneumonia/ or exp Respiratory Tract Infections/ or (asthma* or "chronic obstructive pulmonary disease*" or "chronic obstructive lung disease*" or "chronic obstructive airway disease*" or COPD or COAD or "chronic airflow obstruction*" or "pulmonary emphysema*" or pneumonia* or (pulmonary adj2 inflamm*) or (lung* adj2 inflamm*) or influenza* or flu).ti,ab,kf.
23. exp Blood Pressure/ or exp Hypertension/ or exp Angina Pectoris/ or exp Myocardial Infarction/ or exp Heart Diseases/ or exp Stroke/ or exp Peripheral Arterial Disease/ or ("blood pressure*" or "diastolic pressure*" or "systolic pressure*" or "pulse pressure*" or hypertension* or angina or "myocardial infarct*" or "heart attack*" or "heart disease*" or "cardiac disease*" or stroke or strokes or "cerebrovascular accident*" or CVA or "peripheral arterial disease*" or "peripheral artery disease*").ti,ab,kf.
24. exp Gastrointestinal Diseases/ or exp Inflammatory Bowel Diseases/ or exp Diarrhea/ or exp Enteritis/ or exp Gastroenteritis/ or exp Nausea/ or (gastrointestinal* or "inflammatory bowel*" or diarrhea* or diarrhoea* or enteriti* or gastroenteric* or nausea* or vomiting or emesi).ti,ab,kf.
25. exp Mental Health/ or exp Anxiety/ or exp Depression/ or exp Vertigo/ or exp Stress, Psychological/ or exp Psychological Distress/ or exp Emotions/ or exp Sleep Wake Disorders/ or exp Fatigue/ or exp Headache/ or ("mental health" or anxiety or anxieties or nervousness or anxiousness or depression or depressive or vertigo or vertigos or stress or distress or emotion* or mood or "sleep wake disorder*" or (sleep* adj2 disturb*) or (sleep* adj2 disorder*) or fatigue or headache* or "head pain*").ti,ab,kf.
26. exp Quality of Life/ or Quality-Adjusted Life Years/ or exp Activities of Daily Living/ or Health Status/ or exp Leisure Activities/ or ("quality of life" or (life adj2 quality) or "well being" or wellbeing or (activit* adj2 "daily living") or "outdoor activit*" or "leisure activit*" or QoL or HRQL or HRQoL or WHOQoL or euroqol or "euro qol" or eq5d or "eq 5d" or hql or hqol or "h qol" or hrqol or "hr qol" or hui or hui1 or hui2 or hui3 or "EQ VAS" or "EQ 15D" or "36 Item Short Form Survey" or "SF 36" or "12 item Short Form Survey" or "SF 12").ti,ab,kf.
27. exp Absenteeism/ or ((school* adj2 attend*) or absenteeism).ti,ab,kf.
28. exp Ambulatory Care/ or exp Hospitalization/ or exp Pharmaceutical Preparations/ or exp Anti-Bacterial Agents/ or ("healthcare utilization" or "health care utilization" or ambulatory or outpatient* or "clinic visit*" or "hospital admission*" or hospitalization* or "pharmaceutical preparation*" or medication* or antibiotic* or "anti-bacterial*" or antibacterial* or antiinfective* or "anti infective*").ti,ab,kf.
29. exp Death/ or exp Vital Statistics/ or exp Mortality/ or exp Morbidity/ or (death* or mortalit* or morbidit* or survival or fatal* or "life expectanc*" or "vital statistic*" or "burden of disease" or "disease burden" or "years lived with disability" or YLD or "disability adjusted life" or DALY* or "years of life lost" or YLL or "healthy years equivalent*" or hye or hyes or "healthy days measures" or qaly* or qald* or qale* or qtime* or daly* or "health adjusted life expectancy" or "life years equivalen*" or "well-being adjusted health expectancy").ti,ab,kf.
30. exp "Neoplasms"/ or (neoplas* or cancer* or tumor* or tumour* or malign* or oncolog* or carcinoma*).ti,ab,kf.
31. exp Immune System Diseases/ or exp Autoimmune Diseases/ or exp Arthritis, Rheumatoid/ or exp Psoriasis/ or exp Uveitis/ or exp Hypothyroidism/ or exp Lupus Erythematosus, Systemic/ or exp Multiple Sclerosis/ or exp Spondylitis, Ankylosing/ or ("Immune-mediated disease*" or (immune adj2 disease*) or (immune adj2 disorder*) or (immunologic* adj2 disease*) or (autoimmune adj2 disease*) or (auto-immune adj2 disease*) or (arthritis adj2 rheumatoid) or psorias* or uveitis or uveitide* or hypothyroidism* or hypothyreoidism* or hypothyreosis or hypothyroidea or hypothyroidosis or hypothyrosis or (thyroid* adj2 failure) or (thyroid* adj3 insufficien*) or (thyroid* adj3 deficien*) or "TSH deficien*" or "systemic lupus erythemato*" or "lupus erythematosus disseminatus" or "disseminated lupus" or "erythematodes visceralis" or lupovisceritis or "lupus erythematodes disseminates" or "lupus erythematosus visceralis" or "multiple sclerosis" or "disseminated sclerosis" or "sclerosis disseminated" or "chariot disease" or "insular sclerosis" or "sclerosis

multiplex" or "sclerosis insular" or "sclerosis multiple" or "spondylitis ankylosing" or "ankylosing spondylitis*" or "ankylosing spondylarthriti*" or "ankylosing spondyloarthriti*" or "axial spondyloarthriti*" or "spondylarthriti ankylopoietica" or "spondyloarthriti ankylopoietica" or "rheumatoid spondylitis" or "spondylitis rheumatoid").ti,ab,kf.

32. exp Urinary Tract Infections/ or exp Cystitis/ or exp Vaginitis/ or exp Prostatitis/ or ((genito-urinary adj3 infection*) or (genitourinary adj3 infection*) or ("urogenital tract*" adj3 infection*) or (inflammat* adj3 urogenital*) or ("tractus urogenitalis" adj3 infection*) or (urogenital* adj3 infection*) or ("urinary tract" adj3 infection*) or cystiti* or (bladder adj3 inflammat*) or megacystitis* or pericystitis* or vaginiti* or (vagina* adj3 inflammat*) or colpitis or kolpitis or prostatitis* or (prostate adj3 infection*)).ti,ab,kf.

33. exp Disease Outbreaks/ or exp Endemic Diseases/ or (epidemic* or outbreak* or endemic*).ti,ab,kf.

34. or/16-33

35. "epidemiology".fs. or exp Epidemiologic Methods/ or exp Epidemiologic Studies/ or exp Sentinel Surveillance/ or exp Seroepidemiologic Studies/ or exp Cohort Studies/ or exp Cross-sectional Studies/ or exp Longitudinal Studies/ or exp Follow-up Studies/ or exp Prospective Studies/ or exp Cross-Over Studies/

36. (prevalence or incidence or epidemiol* or survey or rapid assessment or situation assessment or situational assessment or rar or cohort or surveillance or seroprevalence or seroincidence or seroepidemiol* or screening or frequency or occurrence).mp.

37. (((ecological or "follow up" or observational or prospective or cross-over or crossover) adj (studies or study)) or (longitudinal or retrospective or cross-sectional or transversal) or (association adj2 (studies or study))).ti,ab,kf. or (association or associations).ti.

38. or/35-37

39. 9 and 34 and 38

40. 15 or 39

41. exp Animals/ not Humans/

42. (news or editorial or letter).pt.

43. 40 not (41 or 42)

44. remove duplicates from 43

4.6.3. Table 3. Examples of Health Outcomes

Health Outcome Area	Examples
Respiratory tract	• pneumonia • asthma • chronic obstructive pulmonary disease • pulmonary function
Eyes	• conjunctivitis • watery eyes • ocular discharge
Skin	• atopic dermatitis • rash • erythema
Gastrointestinal tract	• diarrhea

	• enteritis • inflammatory bowel disease
Cardiovascular system	• hypertension • angina • myocardial infarction
Genitourinary tract	• genitourinary tract infection • urinary tract infection • cystitis • vaginitis • prostatitis
Infectious Disease	Carriage/colonization of or infection with any of the following pathogens: • <i>Staphylococcus aureus</i> • <i>Coxiella burnetti</i> • <i>Clostridioides difficile</i>
Immune-mediated disease	• rheumatoid arthritis • psoriasis • uveitis • hypothyroidism • hyperthyroidism • systemic lupus erythematosus • multiple sclerosis • ankylosing spondylitis
Cancer	
Mental health	• anxiety • depression • vertigo •
Birth outcomes	• fetal growth retardation • stillbirth • preterm birth •
Healthcare utilization	• medication use • clinic visits • hospitalization
Other	• quality of life

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514
515
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	• absenteeism from work or school • mortality
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